

Linux Cheat Sheet

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Linux File Structure

`/boot`	Boot loader files
`/bin`	executable binaries
`/home`	Users home directory
`/etc`	system config files
`/var/log`	log files
`/usr/local`	software manualy compiled or installed manually

Path shortcute

`~`	home directory
`/`	root directory
`.`	current directory
`..`	parent directory
`-`	previous directory

Linux Terminal

General Commands

`clear`	Clear the terminal screen.
`history`	Display the command history.
`cal`	Display a calendar.
`uptime`	Display how long the system has been running.
`man`	Display the manual for a command. (Tap q to quit the manual)
`date`	Display the current date and time.
`date "+Y-m-d 00:00:00"`	set the date and time.

File Management Commands

`cd`	Change directory.
`ls`	List files in the current directory.
`ls -l`	List files with detailed information.
`ls -a`	List all files, including hidden ones.
`pwd`	Print the current working directory.
`open`	Open a file with the default application.
`mkdir`	Create a new directory.
`rm`	Remove a file.
`rm -rf`	Remove a directory and its contents.
`cp`	Copy a file or directory.
`mv`	Move or rename a file or directory.
`touch`	Create a new empty file.
`cat`	Display the contents of a file.
`tail`	Display the last few lines of a file.
`tail -f`	Continuously display new lines added to a file.
`head`	Display the first few lines of a file.
`stat`	Display detailed information about a file.

OS Commands

`sudo`	Execute a command with superuser privileges.
`sudo -l`	List permitted commands.
`whoami`	Display the current user.
`who`	Display all logged-in users.
`id`	Display user ID and group ID information.
`id -u`	Display the user ID.
`id -g`	Display the group ID.
`ps`	Display currently running processes.
`kill`	Terminate a process by its PID.
`kill -9`	Forcefully terminate a process by its PID.
`reboot`	Restart the system.
`poweroff`	Shut down the system.
`useradd`	Add a new user.
`userdel`	Delete a user.
`passwd`	Change a user's password.
`su - user`	Switch to another user.
`chown user file`	Change the owner of a file to `user`.
`chown user:group file`	Change the owner and group of a file.
`chmod flag file -u tom`	Change the permissions of `file` for user `tom`.
`chmod +x file`	Add execute permission to a file.
`chmod +r file`	Add read permission to a file.
`chmod +w file`	Add write permission to a file.
`chmod -x file`	Remove execute permission from a file.
`chmod -r file`	Remove read permission from a file.
`chmod -w file`	Remove write permission from a file.

Daemons

Systemd

Systemd is the most widely used init system and service manager in modern Linux distributions. It is used in distributions such as Ubuntu, Debian, Fedora, and many others.

`systemctl start service`	Start a service.
`systemctl stop service`	Stop a service.
`systemctl restart service`	Restart a service.
`systemctl status service`	Check the status of a service.
`systemctl enable service`	Enable a service to start on boot.
`systemctl disable service`	Disable a service from starting on boot.
`systemctl daemon-reload`	Reload the systemd manager configuration
`systemctl reset-failed service`	Reset the failed state of a service.
`journalctl -u service`	View logs for a specific service.
`journalctl -n 100 -u service`	View the last 100 log entries for a specific service.
`--no-pager`	Disable paging of log output.

OpenRC

OpenRC is an alternative init system and service manager used in some Linux distributions, such as Alpine Linux and Gentoo.

`rc-service myService start`	Start a service.
`rc-service myService stop`	Stop a service.
`rc-service myService restart`	Restart a service.
`rc-service myService status`	Check the status of a service.
`rc-update add myService default`	Enable a service to start on boot.
`rc-update del myService default`	Disable a service from starting on boot.

Shell Scripting

operators

`>`	Redirect output to a file, overwriting the file if it exists.
`>>`	Redirect output to a file, appending to the file if it exists.
`<`	Redirect input from a file.
`|`	Pipe the output of one command to another command.
`&&`	Execute the next command only if the previous command succeeded.

Basic Syntax

`#!/bin/bash`	Shebang line to specify the script interpreter.
`echo`	Print text to the terminal.
`read`	Read input from the user.
`if`	Start of a conditional statement.
`elif`	Else if condition in a conditional statement.
`else`	Else condition in a conditional statement.
`fi`	End of a conditional statement.
`for`	Start of a for loop.
`while`	Start of a while loop.
`do`	Start of the loop body.
`done`	End of a loop.
`$0`	The name of the script.
`$1, $2, ...`	Positional parameters for script arguments.
`grep`	Search for a pattern in a file or input.
`exec`	Execute a command.
`exit`	Exit the script.

Networking Commands

The below commands requires the `net-tools` package to be installed on the system.

`ifconfig`	Displays IP address.
`ping`	Returns packet loss between two the device and a destination.
`traceroute`	Traces the path packets take to a destination.
`nslookup`	Returns IP address information for a domain name.
`arp -a`	Displays the ARP table, which maps IP addresses to MAC addresses.
`arp myIPa`	Displays the ARP table for the specific IP address.
`curl`	Transfers data from or to a server, using various protocols.
`curl ifconfig.me`	Display machine's public IPv4 address.

SSH connection

`ssh -i ~/.ssh/myKey user@ipv4`	Connect to a remote host via SSH.
`exit`	Exit the SSH session.
`Ctrl + D`	Exit the SSH session.

Cron

Cron is a time-based job scheduler. It allows users to schedule tasks (commands or scripts) to run automatically at specified intervals.

Basics

`crontab -l`	List the current user's crontab.
`crontab -e`	Edit the current user's crontab.
`crontab -r`	Remove the current user's crontab.
`crontab filename`	Install a new crontab from a file.
`crond`	The cron daemon sometime also called `cron`.

Syntax

`* * * * * command`	Run `command` every minute.
`0 * * * * command`	Run `command` at the start of every hour.
`0 0 * * * command`	Run `command` at midnight every day.
`0 0 * * 0 command`	Run `command` at midnight every Sunday.
`0 0 1 * * command`	Run `command` at midnight on the first day of every month
`0 0 1 1 * command`	Run `command` at midnight on January 1st every year.

After the time and date fields, you can specify the command to be executed (note that cron dont use $PATH so you need to use absolute paths). Example:

```
```

This will run the Python script `myscript.py` at midnight every Sunday and reboot the system at 5 AM every day.

package managers

`apt`	Debian-based(include ubuntu) package manager.
`apk`	Alpine Linux package manager.
`yum`	Red Hat-based package manager.
`dnf`	Fedora-based package manager.
`pacman`	Arch Linux package manager.
`brew`	Alternative package manager for macOS and many Linux distributions.

apk

`update`	Update repository indexes.
`upgrade`	Install upgrades available from repositories .
`list`	List all available packages.
`add pkg`	Install a package named `pkg`.
`del pkg`	Remove a package named `pkg`.
`search pkg`	Search for a package named `pkg`.
`info pkg`	Display detailed information about a package named `pkg`.

apt

`update`	Update the package list.
`upgrade`	Upgrade all installed packages to their latest versions.
`list`	List all available packages.
`list --upgradable`	List all packages that can be upgraded.
`install pkg`	Install a package named `pkg`.
`remove pkg`	Remove a package named `pkg`.
`search pkg`	Search for a package named `pkg`.

Tips

Render a website as an app

If a app is not available for linux but can be used on a web browser (eg: outlook, spotify, etc.). you can use the command `chromium-browser --app=siteUrl` (eg: `chromium-browser --app=https://outlook.office365.com/mail`). Create a file /.local/share/applications/outlook.desktop with the following inside:

```
```